Exponential online mistakes for isotropic Bayesian threshold networks at all sufficiently large widths

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

For the fixed discontinuous sign activation, we construct an exactly isotropic Gaussian Bayesian network with exponentially many online mistakes at every sufficiently large hidden width. The model retains one hidden layer, a constant input coordinate, the full Boolean cube, global output margin greater than 1/2, and exact labels. For each $k \geq 2$ the target has a Boolean circuit of size $\Theta(k)$ and a width-one teacher with every weight at most one, while one fixed input permutation forces $2^k - 2$ designated mistakes after an exponential calibration prefix. The result holds under every fixed strong-prior exponent $\gamma > 1$, with any fixed positive variance prefactor, at all sufficiently large widths. It also holds for $\gamma = 1$ with a sufficiently small fixed prefactor. A finite-width Gaussian-tail inequality controls every nonlinear competitor and reduces the posterior to strictly ordered threshold-function cone probabilities. An explicit positive-helper calibration and radial integration produce that order without changing the isotropic covariance. The same target and prior have polynomial IID sample complexity. The cold-posterior width bound is $\mathrm{poly}(k)2^k$; the direct fixed-γ theorem has an additional width requirement stated explicitly. This does not establish the corresponding result for ordinary tanh, ReLU, GELU, or every width above the teacher width.

1 Model and theorem

Let $X = \{-1, 1\}^d$, write $\bar{x} = (1, x)$, and use the fixed activation $\mathrm{sign}(t) = 1$ for $t > 0$ and -1 for $t < 0$. Its value at zero may be fixed arbitrarily: no Gaussian first-layer score is zero on the finite cube with positive probability. The learner has one hidden layer,

$$F_{W,v}(x) = \sum_{a=1}^{N} v_a \,\mathrm{sign}(W_a \cdot \bar{x}), \qquad W_{ai} \sim N(0, N\sigma^2/d), \quad v_a \sim N(0, \sigma^2/N), \tag{1}$$

with all variables independent, including the bias coordinate $i = 0$. There is no coordinate-dependent covariance. For a Boolean function g let

$$U_g = \{g(x)F_{W,v}(x) > 1/2 \text{ for every } x \in X\}. \tag{2}$$

The posterior conditions the prior on the union of these strict global-margin cells and on the exact observed labels. Thus it is a posterior over Boolean functions on the whole cube, not a restriction to threshold subnetworks.

By a posterior error probability at an input we mean the posterior mass of functions assigning the wrong label. Posterior-label majority makes a deterministic mistake whenever this probability

*Authors are listed in alphabetical order.

exceeds 1/2. The stronger threshold 2/3 will also be covered. A fresh posterior function sampled for prediction errs with that posterior probability.

For a fixed $\gamma \geq 1$ and $\tau > 0$, the strong-prior scaling is

$$\sigma_N^2 = \tau^2 N^{-\gamma}, \qquad \mathrm{Var}(W_{ai}) = \frac{\tau^2 N^{1-\gamma}}{d}, \qquad \mathrm{Var}(v_a) = \tau^2 N^{-\gamma-1}. \tag{3}$$

The unit-prefactor version sets $\tau = 1$.

Theorem 1. *For every $k \geq 2$, let $d = 167k + 1617$. There is an explicitly specified threshold function $f_k : X \to \{-1, 1\}$ and a fixed permutation of X with the following properties.*

1. *The target has a bounded-fan-in Boolean circuit of size $\Theta(d) = \Theta(k)$ and a width-one sign-network teacher with every weight at most one in absolute value and output margin one.*

2. *The permutation begins with $2^{d-1}+2$ calibration examples, followed by $2^k - 2$ distinct designated inputs whose true labels are negative. For every fixed $\gamma > 1$, $\tau > 0$, and $\eta > 0$, there is N_0 such that at every $N \geq N_0$ under (3), the posterior error probability at every designated round exceeds $1 - \eta$.*

3. *Consequently posterior-label majority makes all $2^k - 2$ designated mistakes, including under the strict 2/3 criterion, at all sufficiently large widths. The probability that posterior sampling makes all these mistakes can also be made greater than $1 - \eta$ by increasing the same width threshold.*

4. *The conclusion also holds at $\gamma = 1$ for every sufficiently small fixed $\tau > 0$. It is not asserted at $\gamma = 1, \tau = 1$.*

The target, input order, and first-layer isotropy do not depend on N, γ, or τ. The threshold N_0 may depend on k, γ, τ, and the required error allowance.

This answers an all-sufficiently-large-width question for a classical discontinuous activation. It leaves the exactly isotropic tanh question open. The small-anisotropy tanh result of [1] preserves ordinary tanh but changes the covariance; the present result preserves the covariance and changes the activation. The exponential calibration prefix and the large width threshold are substantive features of the construction, not efficient-data or practical-width guarantees.

2 A finite-width posterior bound

Let $\mathcal{H}_d$ be the affine threshold functions on X and let $\pi(g)$ be the probability that one isotropic Gaussian hidden neuron has truth table g. Set $\pi(g) = 0$ for $g \notin \mathcal{H}_d$. For $g \in \mathcal{H}_d$, its strict representation cone is nonempty and open, so $\pi(g) > 0$. Gaussian symmetry gives $\pi(-g) = \pi(g)$. Positive rescaling of every first-layer coefficient does not change the hidden truth table, so these probabilities are independent of the common Gaussian variance and of N.

Write $Q(a) = \mathbb{P}(Z > a)$ for a standard Gaussian Z. Set

$$T = 2^d, \quad c_d = 1 - (1 - 2/T)^2, \quad p_g = 2\pi(g), \quad q_\sigma = Q(1/(2\sigma)), \quad \delta_{N,\sigma} = e^{-c_d/(8N\sigma^2)}. \tag{4}$$

Lemma 2 (All Boolean competitors)**.** *For every Boolean g, every $N \geq 1$, and every $\sigma > 0$,*

$$q_\sigma p_g^N \;\leq\; \mathbb{P}(U_g) \;\leq\; q_\sigma [p_g + (1 - p_g)\delta_{N,\sigma}]^N. \tag{5}$$

Proof. Condition on the hidden truth tables $h_1, \dots, h_N$. Let t count those not equal to either g or $-g$, and put

$$m_a = \frac{1}{T} \sum_{x \in X} g(x) h_a(x).$$

Aligned columns have $|m_a| = 1$. Every other column both agrees and disagrees with g, hence $|m_a| \le 1 - 2/T$. Thus $\|m\|^2 \le N - c_d t$. If U_g occurs, averaging its strict inequalities gives $v \cdot m > 1/2$. This scalar Gaussian has variance $\sigma^2 \|m\|^2 / N$, so

$$\mathbb{P}(U_g \mid h_1, \dots, h_N) \le Q\left(\frac{1}{2\sigma \sqrt{1 - c_d t / N}} \right) \le q_\sigma e^{-c_d t / (8 N \sigma^2)}.$$

For the last inequality use $(1 - r)^{-1} - 1 \ge r$ and $Q(b)/Q(a) \le e^{-(b^2 - a^2)/2}$ for $b \ge a \ge 0$. The latter follows by shifting the Gaussian-tail integral by $b - a$. If all columns are aligned, U_g is exactly one scalar tail of probability q_σ. Alignment occurs independently with probability p_g per neuron. Restricting to $t = 0$ proves the lower bound; averaging the upper bound over its binomial distribution proves the upper bound. □

Corollary 3 (Cold posterior)**.** *At each fixed width N, after any exact-label history consistent with some threshold function, the posterior as $\sigma \downarrow 0$ gives mass proportional to $\pi(g)^N$ on the consistent threshold functions and zero mass on the other Boolean functions.*

Proof. Divide (5) by q_σ and let $\sigma \downarrow 0$. There are finitely many Boolean functions and at least one positive limiting mass. The common factor 2^N cancels on normalization. □

Proposition 4 (Direct strong-prior limit)**.** *Fix any finite collection of exact-label histories. If each has a unique largest-π consistent threshold function, then under (3), for every fixed $\gamma > 1$ and $\tau > 0$, the posterior mass of that function tends to one as $N \to \infty$, simultaneously at all the histories. The same holds for $\gamma = 1$ and all sufficiently small fixed $\tau > 0$.*

Proof. For $\gamma > 1$,

$$\delta_N = \exp[-c_d N^{\gamma - 1} / (8\tau^2)], \qquad N \delta_N \longrightarrow 0.$$

For every threshold function, (5) gives $\mathbb{P}(U_g)/(q_{\sigma_N} p_g^N) \to 1$. For a nonthreshold function it gives $\mathbb{P}(U_g) \le q_{\sigma_N} \delta_N^N$. Dividing by the lower bound for the largest consistent threshold function proves the claim after summing the finitely many other cells.

For $\gamma = 1$, $\delta_N = \delta_\tau = e^{-c_d/(8\tau^2)}$ is constant. At one history let p_* be the largest aligned probability and p_2 the largest among the other consistent threshold functions, taking $p_2 = 0$ if none exist. Include the nonthreshold competitors by the same zero value. Choosing

$$\delta_\tau < \frac{p_* - p_2}{1 - p_2}$$

makes every competing upper-bound base in (5) strictly smaller than the lower-bound base p_*. The finite collection of histories permits one common positive upper bound on τ. □

3 An explicit full-cube calibration

The next construction supplies strictly ordered Gaussian cone probabilities. No parameter is observed or conditioned directly: calibration consists only of exact labels of one threshold function.

3.1 Four registers and positive helper pairs

Fix $k \geq 2$, $p \geq 6$, and $L = p + k + 2$. Consider four logical L-bit registers A, B, C, D, with levels $0, \ldots, L-1$. The actual main inputs are the k bits of A at levels $p, \ldots, p+k-1$. Every other logical bit is represented by a pair of actual helper inputs. There are

$$P = 4L - k \text{ pairs}, \qquad H = 2P = 8L - 2k \text{ helpers}. \tag{6}$$

The signed logical coefficient is $+2^j$ at level j in A, B and -2^j in C, D. Denote the signed coefficient for helper pair l by r_l^*. The canonical main coefficients are $w_i^* = 2^{p+i}$, and put

$$\begin{aligned} R_* &= (2^k - 1)2^p, & S_* &= \sum_l |r_l^*| = 4(2^L - 1) - R_*, \\ H_* &= R_* + S_* + 1 = 2^{L+2} - 3, & u_* &= 1/4. \end{aligned} \tag{7}$$

The two actual helper coefficients of pair l are $H_* + r_l^*/2$ and $H_* - r_l^*/2$. Let $f_0(x, y)$ be the sign of this affine score on the entire cube of main and helper inputs.

All scores are nonzero: each pair contributes either $\pm 2H_*$ or $\pm r_l^*$, so the nonbias score is integral. If the total helper-bit sum $m = \sum y_l$ is nonzero, $|m| \geq 2$, and the label is $\operatorname{sign}(m)$, because $H_* m$ dominates the other terms. If every pair is individually balanced, its inputs are $(s, -s)$ and its contribution is $r_l^* s$. On this subcube the truth table is exactly

$$A + B \geq C + D, \tag{8}$$

where register values are interpreted in 0/1 notation. Equality is positive because $u_* > 0$.

Now consider *any* strict affine representation of the full truth table f_0. Write its main coefficients as w_i, helper coefficients as h_l, and bias as u. Every assertion below applies to this whole representation cone.

3.2 Arithmetic rigidity

Lemma 5. *Let $a = w_0$ and $R = \sum_i w_i$. Then $a > 0$, all $w_i > 0$, and every binary subset sum $A_j = \sum_i \mathrm{bit}_i(j) w_i$, $0 \leq j \leq 2^k - 1$, satisfies*

$$|A_j/a - j| < \epsilon, \qquad \epsilon = 12\, 2^{k-p}. \tag{9}$$

Every within-pair coefficient difference has absolute value at most $10R$, and $|u| \leq 4LR$.

Proof. On the individually pair-balanced subcube, use a_j, b_j, c_j, d_j for the positive register coefficients, writing negative signs for C, D. They are strictly positive because every logical coordinate in (8) is essential and has its indicated monotonicity. In particular $w_i = a_{p+i}$. After switching to 0/1 coordinates and dividing the score by two, it is

$$\beta + \sum_j a_j A_j + \sum_j b_j B_j - \sum_j c_j C_j - \sum_j d_j D_j, \quad \beta = \frac{u - \sum a_j - \sum b_j + \sum c_j + \sum d_j}{2}.$$

The all-zero input and the inputs $C = 1$ or $D = 1$ imply $0 < \beta < \min(c_0, d_0)$. Put $u_0 = \max(c_0, d_0)$. Compare the equality $A = 2^j, C = 2^j$ first with $D = 0$ and then with $D = 1$. The same comparison works with B in place of A, giving

$$-\beta < a_j - c_j < d_0 - \beta, \qquad -\beta < b_j - c_j < d_0 - \beta. \tag{10}$$

Similarly $a_j - d_j, b_j - d_j \in (-\beta, c_0 - \beta)$. All coefficients at a given level differ from a_j by less than u_0, and $|b_j - a_j| < d_0$. For $j < L-1$ compare $A = B = 2^j, C = 2^{j+1}$ first with $D = 0$ and then with $D = 1$. These labels imply

$$\beta - d_0 < c_{j+1} - a_j - b_j < \beta, \qquad |a_{j+1} - 2a_j| < 3u_0. \tag{11}$$

For $j \geq 1$, compare $A = 2^j - 1$ with $C = 2^j$ and with $C = 2^j - 1$. The first label is negative and the second positive, so $c_j > \beta + \sum_{i<j} a_i > \sum_{i<j} c_i$. The same argument gives $d_j > \sum_{i<j} d_i$. Hence

$$a := a_p > (2^{p-1} - 1)u_0 \geq 2^{p-2}u_0.$$

Iterating (11) now gives $|a_{p+i} - 2^i a| < 3u_0(2^i - 1)$. The sum of the k coordinate errors is less than $3u_0 2^k$, which proves (9).

For the coarse coefficient bounds, $u_0 < a/16$ since $p \geq 6$. Above level p, $a_j < (19/16)2^{j-p}a$, while $R > (13/16)(2^k - 1)a$. Thus even at $j = L - 1 = p + k + 1$, $a_j/R < (38/13)2^k/(2^k - 1) \leq 76/13 < 6$. The other register coefficients differ from a_j by less than u_0. Below level p, $c_j < c_p < a + u_0$ and $a_j < c_j + u_0 < a + 2u_0$; all other coefficients there are less than $2R$. Thus $10R$ bounds every register coefficient and every absolute within-pair difference. Finally (10) and its counterpart for $b_j - d_j$ give

$$|u| = \left| 2\beta + \sum_j (a_j - c_j) + \sum_j (b_j - d_j) \right| < (2L + 2)u_0 \leq 4LR.$$

□

3.3 All helpers are positive and have bounded deviations

Lemma 6. *Every helper coefficient satisfies $h_l > R$. Write*

$$H_0 = H^{-1} \sum_l h_l, \qquad \delta_l = h_l - H_0, \qquad \sum_l \delta_l = 0.$$

Then $|\delta_l| \leq 30LR$ for every l.

Proof. Let s_l be the mean of the two coefficients in helper pair l. Fix two distinct pairs i, j, set pair i both positive and pair j both negative, and set every main input positive. Balance all other pairs individually, choosing first all their canonical logical contributions positive and then all negative. Both assignments have total helper-bit sum zero. Their remaining canonical absolute sum is at least

$$S_* - 2\,2^{L-1} = 3\,2^L - R_* - 4 > R_* + 1/4,$$

so the two labels are respectively positive and negative. Arithmetic rigidity fixes the signs of the actual pair differences. If S_{ij} is the sum of their absolute values outside pairs i, j, these two strict inequalities give

$$|2(s_i - s_j) + u + R| < S_{ij} \leq S := \sum_l |h_{l,+} - h_{l,-}| \leq 40LR.$$

Thus $|s_i - s_j| \leq (22L + 1)R$. Averaging over j and adding half a within-pair difference gives $|h_l - H_0| \leq 30LR$.

To prove the stronger positivity statement, fix one actual helper coordinate t. Balance its pair with that coordinate negative, and balance every other pair so that its canonical logical contribution

is negative. The total canonical helper contribution is at most $-S_* + 2\,2^{L-1} < -R_* - 1/4$. Thus the example with all main inputs positive has negative label. Flip only helper coordinate t to positive and put every main input negative. Its total helper-bit sum is now $+2$, so its label is positive. In any fitting representation, the difference of the second score and the first is $2h_t - 2R > 0$. Hence $h_t > R$. □

Lemma 7 (Exact common-mode ray)**.** *Let $\theta = (w, \delta, u)$ with $\sum \delta_l = 0$. The full strict representation cone of f_0 consists exactly of*

$$\theta \in \Omega_0, \qquad H_0 > A(\theta), \tag{12}$$

where Ω_0 is an open homogeneous cone independent of H_0 and

$$R \le A(\theta) \le BR, \qquad F_0 := \sum_i w_i^2 + \sum_l \delta_l^2 + u^2/2 \le B^2R^2, \qquad B = 200L^2. \tag{13}$$

The binary approximation (9) holds on Ω_0.

Proof. Define Ω_0 by the labels on *all* inputs with total helper-bit sum zero, including assignments whose individual pairs are not balanced. Those constraints do not involve H_0. On an input with $m = \sum y_l \neq 0$, the prescribed label is $\mathrm{sign}(m)$; its constraint is $H_0 > -(u + w\cdot x + \delta\cdot y)/m$. Therefore (12) holds with

$$A(\theta) = \max_{x,y:\, m\neq 0} -\frac{u + w\cdot x + \delta\cdot y}{m}.$$

This is a finite maximum, so every $\theta \in \Omega_0$ has a nonempty ray. All pinning and deviation bounds used only total-sum-zero examples and hold on Ω_0. The helper positivity conclusion holds at every point above the ray threshold, hence $A \ge R - \min_l \delta_l \ge R$. Conversely $|m| \ge 2$, $H \le 8L$, and the preceding bounds imply

$$A \le (4L + 1 + 240L^2)R/2 \le 125L^2R.$$

Finally $\sum_i w_i^2 \le R^2$, and $F_0 \le [1 + 7200L^3 + 8L^2]R^2 \le (200L^2)^2R^2$. □

4 Ordering the Gaussian solid angles

This section gives the geometric step of the construction. All Gaussian coordinates in this section have the same variance; by homogeneity we take that variance to be one. The calculation concerns the probability of an *entire threshold function*, rather than the volume of a selected set of its representations.

There are k main bits x, $H \ge 2$ helper bits y, and one selector bit z. Write an affine score as

$$b + cz + \sum_{i=0}^{k-1} w_i x_i + \sum_{\ell=1}^{H} h_\ell y_\ell, \qquad u = b - c, \quad v = b + c.$$

Thus u and v are independent Gaussian variables of variance two, independent of the unit-variance coordinates w_i, h_ℓ. Let $M = 2^k - 1$, and write

$$R = \sum_i w_i, \qquad A_j = \sum_i \mathrm{bit}_i(j) w_i \quad (0 \le j \le M).$$

In particular $A_0 = 0$ and $A_M = R$.

Suppose a complete truth table on the branch $z = -1$ gives an open homogeneous representation cone $\mathcal{C}$ in (w, h, u). We use the following properties of the calibration constructed in the preceding section. Every point of $\mathcal{C}$ has $a = w_0 > 0$, satisfies

$$\left|\frac{A_j}{a} - j\right| < \epsilon \quad (0 \le j \le M), \qquad h_\ell > R \quad (1 \le \ell \le H), \tag{14}$$

and can be parametrized by

$$h_\ell = H_0 + \delta_\ell, \qquad \sum_\ell \delta_\ell = 0, \qquad \theta = (w, \delta, u) \in \Omega_0, \qquad H_0 > A(\theta).$$

Here Ω_0 is an open homogeneous cone, independent of H_0, and A is homogeneous of degree one. For a constant $B \ge 1$,

$$R \le A(\theta) \le BR, \qquad F_0 := \sum_i w_i^2 + \sum_\ell \delta_\ell^2 + u^2/2 \le B^2 R^2. \tag{15}$$

In particular, the condition on H_0 is exactly a lower ray, with no upper restriction. All these statements concern every representation in the calibration cone.

4.1 The compatible functions are exactly the cutoff functions

Let q_j be the point on the branch $z = +1$ whose helper bits are all $+1$ and whose main bits encode the integer j. Add the labels $q_0 \mapsto -1$ and $q_M \mapsto +1$ to the calibration. Assume $\epsilon < 1/2$, so the values A_j increase strictly with j. The negative label at q_0 gives $v + \sum h_\ell - R < 0$. If at least one helper bit is negative, the affine score is at most

$$v + \sum h_\ell - 2 \min_\ell h_\ell + R < 2R - 2 \min_\ell h_\ell < 0.$$

Consequently all threshold functions compatible with the calibration are exactly the M functions $g_0, \dots, g_{M-1}$: they agree with the calibrated function on $z = -1$, are negative off the helper face on $z = +1$, and satisfy

$$g_j(q_i) = -1 \quad (i \le j), \qquad g_j(q_i) = +1 \quad (i > j).$$

Their full representation cones are

$$(w, h, u) \in \mathcal{C}, \qquad -\sum h_\ell + R - 2A_{j+1} < v < -\sum h_\ell + R - 2A_j. \tag{16}$$

Each cell has positive Gaussian measure: $\mathcal{C}$ is nonempty and open, and every displayed interval has positive length. Denote its Gaussian probability by p_j.

4.2 Radial integration and the common helper direction

Set $D = k + H + 1$ and $s = (D + 1)/2$. Normalize $a = w_0$ to one. There are $D + 1$ original affine coefficients, and the radial change of variables has Jacobian a^D. Integrating $a > 0$ in the Gaussian density gives a common positive constant times the power $-s$ of the normalized Gaussian quadratic form. The changes $(b, c) \mapsto (u, v)$ and $h \mapsto (H_0, \delta)$ contribute only constant Jacobians. For fixed normalized θ put

$$\psi_\theta(r) = \int_{A(\theta)}^\infty \left[F_0 + HH_0^2 + \frac{(HH_0 + r)^2}{2}\right]^{-s} dH_0. \tag{17}$$

Equation (16), with $-v = HH_0 + r$, therefore gives

$$p_j = K \int_{\{\theta\in\Omega_0 : w_0 = 1\}} \int_{-R+2A_j}^{-R+2A_{j+1}} \psi_\theta(r)\, dr\, d\theta, \tag{18}$$

where $K > 0$ and the cross-section measure are the same for every j. All inner integration intervals lie in $[-R, R]$.

Define constants

$$\begin{aligned} C_1 &= B^2 + H(B+1)^2 + \frac{[H(B+1)+1]^2}{2}, \\ T_0 &= 2\max\{1, C_1/H\}, \\ C_2 &= B^2 + HT_0^2 + \frac{(HT_0+1)^2}{2}, \\ \ell &= \frac{s(H-1)}{2C_2(M+1)}. \end{aligned}$$

The constants are polynomial in H, B, apart from the explicit factor $(M+1)^{-1}$ in ℓ.

Lemma 8 (A uniform slope after integrating the common direction)**.** *For every normalized θ and every $r \in [-R, R]$, if $\epsilon \le 1$,*

$$-\frac{\psi'_\theta(r)}{\psi_\theta(r)} \ge \ell.$$

Proof. Write $F = F_0 + HH_0^2 + (HH_0 + r)^2/2$. The part of the integral in (17) with $A \le H_0 \le A + R$ is at least $R(C_1R^2)^{-s}$. The part with $H_0 > T_0R$ is at most

$$\int_{T_0R}^{\infty} (HH_0^2)^{-s}\, dH_0 = \frac{H^{-s}(T_0R)^{1-2s}}{2s-1}.$$

The ratio of this upper bound to the preceding lower bound is

$$\frac{(C_1/H)^s T_0^{1-2s}}{2s-1} \le \frac{1}{2}.$$

Indeed $s \ge 1$, and substituting $T_0 = 2\max\{1, C_1/H\}$ proves the inequality separately for $C_1/H \le 1$ and $C_1/H \ge 1$. Thus at least half the probability measure with density proportional to F^{-s} lies in $H_0 \le T_0R$. Differentiation under the integral is justified by the integrable power bounds at infinity and gives

$$-\frac{\psi'_\theta(r)}{\psi_\theta(r)} = s\,\mathbb{E}\left[\frac{HH_0 + r}{F}\right].$$

The integrand is positive everywhere, because $H_0 > A \ge R$ and $H \ge 2$. On $H_0 \le T_0R$, its numerator is at least $(H-1)R$ and its denominator at most C_2R^2. Hence the last display is at least $s(H-1)/(2C_2R)$. Finally (14) gives $R = A_M < M + \epsilon \le M + 1$. □

Proposition 9 (Quantitative ordering of whole function cells)**.** *If*

$$\epsilon \le \frac{1}{8}, \qquad \epsilon \le \frac{\ell}{32},$$

then, for $0 \le j < M - 1$,

$$p_j \ge (1 + \ell/4)p_{j+1}.$$

Proof. For fixed θ, let $l_j = 2(A_{j+1} - A_j)$. By (14), $2 - 4\epsilon < l_j < 2 + 4\epsilon$; in particular $3/2 \le l_j \le 5/2$ and $l_{j+1} - l_j \le 8\epsilon$. Let t be the common boundary of the two adjacent integration intervals. Lemma 8 implies

$$\int_{t-l_j}^{t} \psi_\theta(r)\, dr \ge \psi_\theta(t)(l_j + \ell l_j^2/2), \qquad \int_{t}^{t+l_{j+1}} \psi_\theta(r)\, dr \le \psi_\theta(t) l_{j+1}.$$

The difference of the first mass and the second is at least

$$\psi_\theta(t)(\ell l_j^2/2 - 8\epsilon) \ge (7\ell/8)\psi_\theta(t),$$

whereas the second mass is at most $(5/2)\psi_\theta(t)$. Thus the first is at least $(1 + 7\ell/20)$ times the second, which is stronger than the asserted factor $1 + \ell/4$. Integrate this inequality in (18). □

4.3 A quantitative width bound in the cold posterior

Take the target to be g_{M-1} and, after calibration, reveal $q_1, q_2, \ldots, q_{M-1}$, each with label -1. There are $T = M - 1 = 2^k - 2$ designated rounds. Suppose the cold posterior on surviving threshold functions is proportional to p_j^N, as established by the posterior calculation for sign networks. Immediately before q_i is revealed, the survivors are $g_{i-1}, \ldots, g_{M-1}$. Exactly g_{i-1} predicts the wrong label $+1$. Writing $\rho = 1 + \ell/4$, Proposition 9 and a geometric-series bound give

$$\mathbb{P}(\text{correct label at } q_i \mid \text{past labels}) \le \rho^{-N}. \tag{19}$$

For completeness, if the surviving weights are $b_0, b_1, \ldots$ with $b_{r+1} \le q b_r$ and $q = \rho^{-N}$, their tail has mass at most $q b_0/(1-q)$, so its posterior fraction is at most q. Consequently

$$N \ge \frac{\log(T/\eta)}{\log(1 + \ell/4)}$$

ensures, by a union bound, that sampled posterior predictions make every one of these T mistakes with probability at least $1 - \eta$. The same estimate implies a wrong posterior majority at every designated round whenever $\rho^{-N} < 1/2$. When H and B are polynomial in k, the displayed cold width threshold is $2^k \,\mathrm{poly}(k) \log(T/\eta)$. This quantitative threshold concerns the cold posterior. A simultaneous width-and-scale limit also needs the separate finite-scale posterior error bound; its dimension-dependent constants should not be suppressed into this cold width estimate.

5 Completing the construction and the width statements

5.1 An explicit precision choice

Set $p = 20k + 200$ and $L = 21k + 202$. There are $k + H$ base inputs and one selector, so $d = k + H + 1 = 167k + 1617$. The constants in the volume argument satisfy

$$C_1 \le 7H^2B^2, \qquad T_0 \le 14HB^2, \qquad C_2 \le 784H^4B^4.$$

Consequently its logarithmic-slope bound satisfies

$$\ell \ge \frac{1}{1568 H^4 B^4\, 2^k} > 2^{-54} L^{-12} 2^{-k}. \tag{20}$$

The required inequalities $\epsilon < 1/8$ and $\epsilon < \ell/32$ hold: the latter follows from $p > 2k + 63 + 12\log_2 L$. Indeed $L \le 223k$, so $\log_2 L < 8 + \log_2 k \le k + 7$, and $p = 20k + 200 > 14k + 147$.

5.2 The target and the fixed permutation

Use the canonical main/helper coefficients from (7) and the branch bias $u_* = 1/4$ on selector $z = -1$. Put $H_\Sigma = \sum_l h_l$ and $a = 2^p$. On the branch $z = +1$ choose

$$v_* = -H_\Sigma - R_* + a.$$

The two branch biases are implemented by the ordinary constant and selector coefficients $b = (u_* + v_*)/2$ and $c = (v_* - u_*)/2$. On the all-positive helper face, the all-positive main input has score $a > 0$ and every other main input has score at most $-a < 0$. Every off-face input is negative by the helper bound. Thus the full target is f_0 on the first branch and the AND of all main/helper bits on the second branch. It is exactly the last finite cutoff function in the volume construction.

Take the fixed input permutation to be: all $z = -1$ inputs in lexicographic order; then the two face endpoints q_0, q_M where $M = 2^k - 1$; then $q_1, \ldots, q_{M-1}$; then all remaining inputs in lexicographic order. These blocks are disjoint. The first endpoint is negative, the last positive, and every designated middle input is negative. After $q_1, \ldots, q_j$ have been observed, the consistent threshold functions are exactly the cutoff functions indexed $j, j+1, \ldots, M-1$. The first has strictly largest cone probability and predicts q_{j+1} positively. There are exactly $M - 1 = 2^k - 2$ such rounds.

Proposition 4 proves the posterior statement of Theorem 1. For posterior sampling, apply that statement with per-round error allowance $\eta/(M-1)$ and use the union bound; no independence between predictions is required.

5.3 Teacher and Boolean circuit complexity

The canonical full threshold representation has no zero score on the cube. Multiply all its first-layer coefficients, including the bias, by a sufficiently small positive scalar so that their absolute values are at most one. The hidden sign function is unchanged. Give the single hidden neuron output weight one. This is the claimed width-one, weight-one teacher, with global output margin one.

For completeness the target also has a linear-size Boolean circuit; the bounded teacher is not the only complexity guarantee. Count the $H = O(L)$ helper bits using carry-save addition. Grouping three bits of one weight into a full adder reduces the number of live bits by one, so $O(H)$ constant-size full adders suffice, followed by an $O(\log H)$ ripple addition. The common coefficient $H_* = 2^{L+2} - 3$ multiplies that count by a shift and two subtractions, using $O(L + \log H)$ gates. After conversion of signs to 0/1 bits, the remaining dyadic terms contain at most a constant number of source bits per level: at most eight from the four helper-pair registers and the main bits that replace one of those pairs. Sum positive and negative terms separately with carry-save addition and subtract the results. This requires $O(L)$ gates and $O(L)$-bit arithmetic. Include the fixed bias and compare the sign. This computes f_0 in $O(L)$ gates. A linear size AND and a selector multiplexer compute the other branch and the full target. Conversely every main/helper input is essential through the AND branch. A bounded-fan-in circuit depending on $d - 1$ distinct inputs needs $\Omega(d)$ gates. Since $L = \Theta(d) = \Theta(k)$, the circuit size is $\Theta(d)$.

5.4 What the width bound means

The quantitative adjacent cone ratio from the volume argument gives a cold-posterior threshold of $\mathrm{poly}(k)2^k$ for any fixed confidence. A logarithmic confidence factor suffices. This means that at every width above that threshold, taking the scalar scale σ sufficiently small produces the designated mistakes.

Under a prescribed $\gamma > 1$ and $\tau > 0$, one must additionally make

$$N \exp[-c_d N^{\gamma-1}/(8\tau^2)]$$

sufficiently small relative to the finite cone probabilities, as well as suppress the nonthreshold mass in (5). These are explicit sufficient inequalities, and hold eventually for every fixed $\gamma > 1$. Since c_d is of order 2^{-d}, the resulting fixed-γ width threshold need not be $\text{poly}(k)2^k$. No lower bound at every width $N \geq 1$ or at every width above the width-one teacher is claimed.

6 Polynomial IID sample complexity for the same prior

The lower bound is specific to adversarial ordering. The same actual posterior has a polynomial realizable IID guarantee for this target.

Theorem 10. *Fix any distribution D on X, a threshold target f, and numbers $\varepsilon, \beta \in (0,1)$. If*

$$m \geq \frac{4}{\varepsilon}[(d+1)^2 \log 2 + \log(1/\beta)], \tag{21}$$

then, with probability at least $1-\beta$ over m IID labeled examples, posterior-label majority under (3) has D-error at most ε at every sufficiently large width, for any fixed $\gamma > 1$ and $\tau > 0$. The width threshold can be chosen independently of the training sample and D. The assertion also holds at $\gamma = 1$ for every sufficiently small fixed $\tau > 0$.

Proof. The $T = 2^d$ input points give T hyperplanes in the $(d+1)$-dimensional coefficient space. The usual hyperplane-region recurrence, obtained by adding one hyperplane at a time, bounds their number of cells by $\sum_{j=0}^{d+1} \binom{T}{j} \leq (T+1)^{d+1} \leq (2T)^{d+1}$. Every strict threshold truth table is one such cell. Hence $\log |\mathcal{H}_d| \leq (d+1)^2 \log 2$. A union bound shows that the probability any consistent threshold function has D-error greater than $\varepsilon/4$ is at most $|\mathcal{H}_d| e^{-m\varepsilon/4} \leq \beta$.

The target is always consistent. Lemma 2 bounds the posterior mass of all nonthreshold functions, uniformly over every target-consistent history, by

$$B_N = \min\left\{1,\ 2^T \left(\frac{\delta_N}{p_f}\right)^N\right\}. \tag{22}$$

For fixed $\gamma > 1$, $B_N \to 0$. For $\gamma = 1$, choosing $\delta_\tau < p_f$ also gives $B_N \to 0$. On the high-probability sample event, the posterior mean of population error is at most $\varepsilon/4 + B_N$. The population error of posterior label majority is at most twice that mean, because a wrong majority puts at least half the posterior mass on the wrong label. Once $B_N \leq \varepsilon/4$, the asserted bound follows. □

For the explicit target, $d = \Theta(k)$, so the IID sample bound is $O((k^2 + \log(1/\beta))/\varepsilon)$, while the adversarial order forces $2^k - 2$ mistakes. One sufficiently small fixed prefactor at $\gamma = 1$ can satisfy both the online and IID conclusions by taking the minimum of their positive bounds. The theorem concerns sample complexity; the necessary learner width and posterior computation are not claimed to be polynomial.

7 Scope

All labels used above are labels of one explicit affine threshold function on the full input cube. The calibration does not pin parameters inside a surviving truth-table cell: it eliminates incompatible

cells, and the remaining cells have different Gaussian solid angles. The posterior comparison includes every Boolean output obtainable by the wide sign network, including outputs that are not thresholds. No unfavorable choice from an isotropic tie is made.

The activation is discontinuous sign. The result therefore does not transfer to standard tanh by approximation: an exponentially narrow threshold cone and an unbounded width limit would require new uniform control. The exactly isotropic all-width lower-bound questions for ordinary tanh, ReLU, and GELU remain outside this theorem.

References

[1] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Exponential online mistakes at every width after arbitrarily small Gaussian anisotropy*. 2026. Companion manuscript.